

Computing, and Data Science. The framework classifies candidate proficiency levels into Beginner, Intermediate, and Advanced categories based on competency density, contextual project exposure, implementation complexity, and semantic relevance.

The proposed platform further incorporates a semantic job-role compatibility engine capable of comparing unified competency embeddings with industrial job-role embeddings using cosine similarity metrics and semantic matching algorithms. Unlike traditional ATS systems that depend on exact keyword overlap, the proposed semantic similarity framework performs contextual employability analysis capable of identifying latent competency relationships and hidden professional alignments.

Another major component of the proposed framework is the intelligent skill gap analysis engine. The system identifies missing competencies and categorizes deficiencies into Critical Skill Gaps, Moderate Skill Gaps, and Optional Skill Enhancements. Based on these analytical outputs, the adaptive recommendation engine dynamically generates personalized learning roadmaps, certification suggestions, coding practice strategies, project recommendations, and competency enhancement plans tailored to individual career objectives and domain requirements.

To ensure transparency, interpretability, and fairness, the proposed framework integrates Explainable Artificial Intelligence (XAI) techniques such as SHAP and LIME into the employability prediction pipeline. These mechanisms provide interpretable explanations for competency scores, recommendation outcomes, feature importance analysis, and employability classifications. The Explainable AI layer enables students, recruiters, and institutional administrators to understand the reasoning behind AI generated decisions and recommendation outputs.

The proposed system also includes an interactive employability analytics dashboard capable of visualizing competency scores, employability readiness, domain wise proficiency levels, recommendation analytics, skill gap distributions, and career progression metrics. The dashboard supports intelligent decision making for students, educational institutions, placement departments, and recruitment agencies through real time analytical visualization and competency monitoring.

Furthermore, the proposed framework is designed with modularity, scalability, extensibility, and research adaptability as core architectural principles. The architecture supports integration with cloud deployment environments, institutional recruitment infrastructures, coding assessment

platforms, and intelligent educational analytics systems. The framework can also be extended toward AI driven interview intelligence, multimodal competency analysis, GitHub repository evaluation, and real-time labor market analytics integration.

Therefore, the proposed SkillNova AI framework provides a comprehensive semantic competency intelligence ecosystem capable of delivering contextual employability analysis, adaptive recommendation generation, explainable AI driven career guidance, unified competency profiling, and intelligent employability enhancement through a unified AI driven competency enhancement framework.

1.5.1 Existing System Vs Proposed System

Sl. No	Feature	Existing System	Proposed System
1	Resume Analysis	Performs traditional keyword-based resume screening using lexical matching and rule-based filtering mechanisms	Performs semantic resume understanding using advanced NLP and transformer-based architectures such as BERT and Sentence-BERT
2	Skill Extraction	Extracts only explicit keywords and predefined technical terms from resumes	Performs contextual extraction of technical competencies, certifications, projects, frameworks, and domain expertise using NLP and Named Entity Recognition
3	Semantic Understanding	Lacks contextual interpretation and semantic relationship modeling capabilities	Utilizes contextual embedding techniques and transformer-based semantic representations for deep competency understanding
4	Competency Profiling	Evaluates candidate skills using binary or static classification approaches	Performs multidimensional competency profiling with domain-wise proficiency classification and contextual competency evaluation
5	Employability Prediction	Provides limited employability assessment with low analytical intelligence	Generates AI-driven employability prediction scores using hybrid Machine Learning and Deep Learning models

6	Job Matching	Relies on exact keyword overlap between resumes and job descriptions	Uses semantic similarity computation and contextual embedding-based job-role compatibility analysis
7	Skill-Gap Analysis	Unable to effectively identify hidden competency deficiencies and learning gaps	Performs intelligent skill-gap detection and categorization using semantic competency comparison techniques
8	Recommendation System	Generates static and generalized recommendations without personalization	Produces adaptive and personalized learning roadmaps, certification recommendations, and competency enhancement strategies
9	Competency Data Integration	Academic records, coding profiles, certifications, and project data remain fragmented and isolated	Integrates heterogeneous competency sources into a unified semantic competency intelligence framework
10	Scalability	Faces limitations in handling large-scale semantic analysis and multidomain competency evaluation	Designed as a scalable AI-driven architecture capable of supporting large-scale employability analytics and intelligent recommendation generation

1.5.2 Advantages of Proposed System

The proposed SkillNova AI framework offers several significant advantages over conventional employability analysis and recruitment systems by integrating semantic intelligence, adaptive recommendation mechanisms, Explainable Artificial Intelligence, and unified competency analytics into a scalable AI driven architecture. The system is designed to improve employability prediction accuracy, recommendation quality, competency understanding, and intelligent career enhancement through advanced Natural Language Processing technologies.

One of the major advantages of the proposed system is its capability to perform semantic resume understanding using transformer based NLP models such as BERT and Sentence BERT. Unlike traditional keyword-based Applicant Tracking Systems that depend on lexical matching, the proposed framework performs contextual competency analysis capable of understanding semantic relationships between technical skills, frameworks, projects, certifications, and industrial job requirements. This significantly improves recommendation relevance and reduces false candidate